

balances of neurotransmitters like dopamine and serotonin are biologically or genetically set. Then, other influences make them more or less intense’.

In other words, while nature might bring about the inclination to spend or save, the relationship between nature and nurture affects whether we *act* on that inclination.

Why do biochemical balances matter when it comes to money? A 2005 research paper titled ‘Neural predictors of purchases’ by Brian Knutson, Scott Rick, G. Elliott Winmer, Drazen Prelec and George Loewenstein studied the brain’s reactions when consumers made purchasing decisions. What they already knew and acknowledged was that other researchers had already been studying how the brain reacted to purchasing decisions. For example, the brain reacts to our product preferences with research proving that men who looked at pictures of sports car versus a regular sedan showing increased midbrain MPFC activity. What Knutson and his fellow researchers in this study found was that pleasure centres in the brain were activated when some consumers anticipated a purchase because they were about to receive a gain. While pain centres in the brain were activated when some consumers believed that the price was excessively high because they were experiencing a loss.

In other words, the researchers could tell through changes in biochemical balances and activation in the brain that the consumers were experiencing real feelings of pleasure and pain when it came to their spending choices. There was real emotion attached to these choices and depending on the level of brain activation, this affected purchasing decisions.

But does this mean we’re hardwired, depending on the level of pleasure and pain we experience, to be a penny pincher or a big spender? In a 2008 study called ‘Tightwads and Spendthrifts’, George Loewenstein found that different people experience different levels of ‘pain paying’. He concluded this meant, depending on the degree of pain experienced, that people were either more a ‘tightwad’ or a ‘spendthrift’. Loewenstein has commented that he believed ‘it is unlikely that people will change from being a big spender to being a big saver or vice versa ... perhaps it is genetic.’

However, more and more researchers, including Anne Mostafa, argue that while we have this inclination towards spending or saving behaviour,